



**AN EXPLAINABLE ATTENTION-BASED
MULTIMODAL DEEP LEARNING
FRAMEWORK FOR ACTIVE AND
LATENT TUBERCULOSIS DETECTION**

A PROJECT REPORT

Submitted by

SANTHOSH T	811722104134
SWATHI R	811722104165
VIGNESH V	811722104180
VIGNESHWARAN R	811722104181

in partial fulfillment of the requirements for the award degree of

Bachelor of Engineering

20CS8504 - PROJECT WORK

**DEPARTMENT OF COMPUTER SCIENCE AND
ENGINEERING**

K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY

(AUTONOMOUS)

SAMAYAPURAM - 621112

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BONAFIDE CERTIFICATE

The work embodied in the present project report entitled “**AN EXPLAINABLE ATTENTION-BASED MULTIMODAL DEEP LEARNING FRAMEWORK FOR ACTIVE AND LATENT TUBERCULOSIS DETECTION**” has been carried out by the students **SANTHOSH T, SWATHI R, VIGNESH V, VIGNESHWARAN R**. The work reported herein is original and we declare that the project is their own work, except where specifically acknowledged, and has not been copied from other sources or been previously submitted for assessment.

Date of Viva Voce:

Mrs. R. Deepa, M.E.,
SUPERVISOR
Assistant Professor
Department of CSE
K. Ramakrishnan College of
Technology
Samayapuram- 621112

Mr. R. Rajavarman, M.E,(Ph.D)
HEAD OF THE DEPARTMENT
Assistant Professor(Sr. Grade)
Department of CSE
K. Ramakrishnan College of
Technology
Samayapuram- 621112

INTERNAL EXAMINER

EXTERNAL EXAMNIER

ABSTRACT

In modern healthcare, early detection of tuberculosis is essential for improving patient outcomes, yet access to expert radiological interpretation remains limited, especially in resource-constrained environments. This project presents a Multimodal Tuberculosis Detection System, an intelligent chest X-ray analysis and diagnostic assistance platform powered by deep learning and explainable artificial intelligence. The system enables automated detection of tuberculosis through a user-friendly web interface, providing fast, reliable, and accessible diagnostic insights without requiring specialized clinical expertise. The platform utilizes a CNN-based deep learning model to analyze chest X-ray images, identify pulmonary abnormalities, assess severity levels, and classify the infection as Active or Latent Tuberculosis based on clinical risk parameters. Grad-CAM is integrated to generate visual heatmaps that highlight the lung regions influencing the model's decision, ensuring transparency and clinical trust. The system incorporates multimodal clinical data analysis using CBC, ESR, and CRP values, automated PDF diagnostic report generation, and secure patient history management. Its backend is built using the Flask framework with SQLite database support, ensuring reliable data handling, while the frontend delivers an intuitive experience for patients and healthcare professionals alike.

Keywords: Tuberculosis Detection, Chest X-Ray Analysis, Deep Learning, CNN, Explainable AI, Grad-CAM

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SIGNATURE

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LIST OF ABBREVIATIONS

AI	-	Artificial Intelligence
CNN	-	Convolutional Neural Network
TB	-	Tuberculosis
XAI	-	Explainable Artificial Intelligence
Grad-CAM	-	Gradient-weighted Class Activation Mapping
MRI	-	Magnetic Resonance Imaging
CT	-	Computed Tomography
CBC	-	Complete Blood Count
ESR	-	Erythrocyte Sedimentation Rate
CRP	-	C-Reactive Protein
API	-	Application Programming Interface
NLP	-	Natural Language Processing
SVM	-	Support Vector Machine
ANN	-	Artificial Neural Network
KNN	-	K-Nearest Neighbors
JWT	-	JSON Web Token
GLCM	-	Gray Level Co-occurrence Matrix
LBP	-	Local Binary Pattern
ViT	-	Vision Transformer

CHAPTER 1

INTRODUCTION

1.1 DESCRIPTION

Tuberculosis (TB) is one of the most serious infectious diseases affecting millions of people worldwide. It is caused by the bacterium *Mycobacterium tuberculosis*, which primarily affects the lungs but can also impact other parts of the body. According to the World Health Organization (WHO), tuberculosis remains one of the leading causes of death globally, particularly in developing countries where healthcare infrastructure may be limited. Early detection and accurate diagnosis are essential for effective treatment and to prevent the spread of the disease.

Traditional methods for TB diagnosis include chest X-rays, sputum smear microscopy, and laboratory tests such as CBC (Complete Blood Count), ESR (Erythrocyte Sedimentation Rate), and CRP (C-Reactive Protein). While these methods are effective, they often require expert medical professionals and laboratory facilities. In many regions, especially rural or underdeveloped areas, access to such medical resources is limited. This leads to delays in diagnosis and treatment, which can worsen the patient's condition and increase the risk of disease transmission.

With the rapid advancement of artificial intelligence (AI) and machine learning technologies, automated medical diagnostic systems are becoming increasingly popular. AI-based healthcare systems can analyze large volumes of medical data quickly and provide accurate predictions. In particular, deep learning models such as Convolutional Neural Networks (CNNs) have shown remarkable success in medical image analysis. These models can automatically extract patterns from medical images and classify diseases with high accuracy.

This project proposes a Multimodal Tuberculosis Detection System that integrates medical imaging data and clinical parameters to provide accurate TB diagnosis. The system uses a CNN model trained on chest X-ray images to detect whether tuberculosis is present or not. Additionally, clinical data such as CBC, ESR, and CRP values are used to estimate the severity or risk level of the disease.

Another important aspect of modern AI systems is Explainable Artificial Intelligence (XAI). Medical professionals often require explanations for AI predictions before trusting the system. Therefore, this project integrates Grad-CAM (Gradient-weighted Class Activation Mapping) to visualize the regions in the chest X-ray image that influenced the AI model's decision. This provides transparency and helps doctors understand the reasoning behind the prediction.

The system also includes additional functionalities such as user authentication, patient data management, prediction history tracking, and automatic generation of diagnostic reports in PDF format. The user interface is designed using a web-based platform, making it easy for healthcare professionals to interact with the system.

Overall, the proposed system aims to assist doctors and healthcare workers by providing an intelligent decision-support tool for tuberculosis detection. By combining deep learning, clinical analysis, and explainable AI, the system can improve diagnostic accuracy and contribute to early detection and better management of tuberculosis.

1.2 DOMAIN SPECIFICATION

1.2.1 Artificial Intelligence in Healthcare

Artificial Intelligence has revolutionized many industries, and healthcare is one of the most significant domains benefiting from its applications. AI technologies are capable of analyzing complex medical data, detecting patterns, and assisting doctors in diagnosing diseases. In recent years, machine learning and deep learning algorithms have been widely used for disease detection, medical image analysis, and predictive healthcare analytics.

AI-based diagnostic systems can process medical images such as X-rays, CT scans, and MRI images with high accuracy. These systems can identify abnormalities that might be difficult for the human eye to detect. Moreover, AI systems can operate continuously without fatigue, making them useful for assisting doctors in busy healthcare environments.

In the context of tuberculosis detection, AI can analyze chest X-ray images to identify patterns associated with TB infection. When combined with clinical data such as blood test results, AI systems can provide more accurate and reliable predictions. This integration of multiple data sources is known as multimodal analysis, which improves diagnostic performance.

Another important advantage of AI in healthcare is its ability to provide decision support. Rather than replacing doctors, AI systems act as intelligent assistants that help medical professionals make informed decisions. By analyzing patient data quickly and accurately, AI systems can reduce diagnosis time and improve patient outcomes.

1.2.2 Deep Learning and Convolutional Neural Networks

Deep learning is a subset of machine learning that uses artificial neural networks with multiple layers to learn complex representations from data. Among the various deep learning architectures, Convolutional Neural Networks (CNNs) are particularly effective for image analysis tasks.

CNNs are designed to automatically extract features from images using convolutional layers, pooling layers, and fully connected layers. Unlike traditional machine learning methods that require manual feature extraction, CNNs learn important image features directly from the data during training. This makes them highly effective for tasks such as object detection, image classification, and medical image analysis.

In medical imaging, CNNs have been successfully applied to detect diseases such as pneumonia, cancer, diabetic retinopathy, and tuberculosis. By training on large datasets of labeled medical images, CNN models can learn patterns that distinguish healthy tissues from diseased tissues.

In this project, a CNN model based on the ResNet architecture is used to analyze chest X-ray images. The model processes the image and predicts whether tuberculosis is present or not. This automated analysis helps in early disease detection and reduces the workload of medical professionals.

1.2.3 Explainable Artificial Intelligence

One of the major challenges of AI systems is the lack of transparency in their decision-making process. Deep learning models are often referred to as "black boxes" because it is difficult to understand how they arrive at their predictions. In critical domains such as healthcare, this lack of transparency can reduce trust in AI systems.

Explainable Artificial Intelligence (XAI) addresses this problem by providing methods that explain how AI models make decisions. These explanations help doctors understand the reasoning behind AI predictions and verify whether the model's conclusions are medically valid.

In this project, Grad-CAM is used as the explainability technique. Grad-CAM generates a heatmap over the chest X-ray image to highlight the regions that influenced the model's prediction. For example, if the model detects tuberculosis, the heatmap will highlight the lung regions where abnormalities were identified. This visual explanation provides valuable insights for doctors and increases confidence in the AI system's predictions.

1.3 OBJECTIVES OF THE STUDY

The primary objective of the Multimodal Tuberculosis Detection System is to develop an intelligent, AI-powered diagnostic assistant capable of analyzing chest X-ray images, generating medical insights, and guiding patients toward appropriate healthcare decisions.

Specific objectives include:

- To build an automated chest X-ray analysis system using CNN-based deep learning for accurate tuberculosis detection and severity classification.
- To implement a secure patient authentication workflow for managing user profiles and medical history.
- To generate structured, radiology-style diagnostic reports using AI, including tuberculosis presence, severity, and medical explanations.
- To enable Grad-CAM-based visual explanations, allowing healthcare professionals to understand AI predictions.
- To integrate clinical risk analysis using CBC, ESR, and CRP values for a comprehensive multimodal diagnosis.
- To provide an automated PDF report generation module for medical documentation.
- To ensure a user-friendly, responsive web interface using Flask, making it accessible for healthcare professionals.
- To maintain high data privacy and security standards, ensuring that sensitive patient information is protected.

1.4 SCOPE OF THE PROJECT

The Multimodal Tuberculosis Detection System is designed for individuals and healthcare providers seeking an intelligent, accessible, and accurate system for early tuberculosis detection and medical guidance. It is ideal for:

- Patients who want fast, AI-assisted preliminary analysis of their chest X-ray images.
- Doctors and Radiologists who need AI-generated summaries to support diagnostic decisions.
- Medical Students who want an interactive tool to learn X-ray interpretation.

- Telemedicine Platforms offering remote diagnostic assistance and specialist recommendations.

1.4.1 Functional Scope

The AI-powered Tuberculosis Detection System integrates multiple smart healthcare modules into one platform, beginning with an advanced Chest X-Ray Interpretation Module that uses CNN-based deep learning to analyze uploaded images, detect tuberculosis presence, assess severity, and identify structural abnormalities. It includes a secure Patient Authentication Module for managing user accounts, medical history, and previous diagnostic results. The system automatically produces structured diagnostic reports containing tuberculosis status, severity level, recommended specialists, AI-generated explanations, and scan metadata. A built-in Clinical Risk Analysis module estimates the risk level based on CBC, ESR, and CRP values, while the Patient History Dashboard tracks past diagnoses and severity trends. A PDF Report Export option allows generating downloadable medical reports for clinical or personal use.

1.4.2 Security and Data Handling Scope

The system incorporates strong security features, including secure authentication mechanisms to ensure that only authorized users can view or modify personal medical data. All sensitive information such as patient profiles, medical history, and diagnostic results is stored securely in the database, preventing unauthorized access. The platform enforces a strict no-sharing policy, ensuring that X-ray images and health data are never shared with third parties, and remain entirely within the system unless the user chooses to download a report. It also adheres to medical privacy principles through a privacy-by-design approach, using industry-standard encryption to safeguard all health information. Additionally, the system performs AI processing without long-term storage of uploaded images, meaning that images are analyzed in real time and are not saved externally.

1.5 SIGNIFICANCE OF THE PROJECT

The Multimodal Tuberculosis Detection System significantly advances the field of medical diagnostic technology by combining artificial intelligence with accessible digital healthcare. It empowers individuals to obtain rapid, AI-assisted X-ray interpretations, reducing anxiety and improving early awareness of potential pulmonary conditions. By providing structured analysis, severity grading, and clinical risk assessment, the system enhances patient understanding and supports timely medical action.

The project's use of advanced CNN models and Grad-CAM explainability demonstrates a powerful shift toward intelligent, assistive diagnostic tools that do not replace medical professionals but strengthen decision-making and patient engagement. Its secure data handling, intuitive interface, and automated reporting features make it a reliable resource for both patients and healthcare practitioners. Ultimately, the system contributes to earlier detection, improved patient literacy, and more informed clinical pathways, marking it as an important step toward accessible, AI-enhanced medical support systems that prioritize accuracy, security, and user empowerment.

CHAPTER 2

LITERATURE SURVEY

2.1 DEEP LEARNING APPROACHES FOR TUBERCULOSIS DETECTION IN CHEST X-RAYS

A. Sharma et al. in their research explain that the rapid evolution of deep learning technologies in 2023 significantly transformed medical image analysis, particularly for tuberculosis detection using chest X-ray images. Traditional diagnostic approaches relied heavily on manual interpretation by radiologists, which could be time-consuming and prone to human error. The introduction of advanced Convolutional Neural Network (CNN) architectures such as ResNet, DenseNet, and EfficientNet enabled automated systems to detect tuberculosis-related abnormalities with high precision. These models are capable of learning complex visual features directly from chest X-ray images without the need for manual feature extraction.

Deep learning models trained on large-scale radiological datasets demonstrated strong capability in identifying tuberculosis patterns such as lung cavities, infiltrates, nodules, and consolidation areas. Researchers highlighted that transfer learning techniques played a crucial role in improving model performance by leveraging pre-trained models originally developed on large image datasets. This approach significantly reduced training time while improving classification accuracy. Furthermore, studies conducted in 2023 emphasized the importance of domain-specific training and medical dataset fine-tuning to ensure clinical reliability and safety. Several evaluations showed that CNN-based models achieved diagnostic performance comparable to trained radiologists in distinguishing tuberculosis-positive cases from normal lung structures. Another major advancement discussed in the study was the integration of Grad-CAM visualization techniques. Grad-CAM heatmaps provide visual explanations by highlighting the lung regions that influence the model's prediction. This capability allows radiologists to verify whether the AI system is focusing on medically relevant areas of the X-ray image.

2.2 ENHANCED PERFORMANCE OF DEEP LEARNING MODELS FOR TUBERCULOSIS CLASSIFICATION

L. Wang et al. describe that the year 2022 witnessed major progress in deep learning-based tuberculosis classification systems, particularly with the widespread adoption of advanced convolutional neural network architectures and hybrid deep learning frameworks. These models significantly improved the ability of automated systems to identify tuberculosis-related abnormalities in chest X-ray images. Researchers experimented with high-performance architectures such as EfficientNet-B7, DenseNet-201, and MobileNet-V3, which demonstrated superior performance in feature extraction and classification tasks.

These deep learning architectures are capable of learning hierarchical features from chest X-ray images. Early layers detect basic image patterns such as edges and textures, while deeper layers identify more complex patterns associated with pulmonary abnormalities, including lesions, cavities, and irregular lung opacities. As a result, these models achieved improved diagnostic accuracy compared to earlier machine learning techniques.

A key focus of research during this period was improving model generalization and preventing overfitting. Researchers implemented data augmentation techniques such as image rotation, flipping, brightness adjustment, and scaling to increase dataset diversity. Additionally, optimization techniques including Ranger optimizer, AdamW optimizer, dropout layers, and batch normalization were widely adopted to improve training stability and model performance.

Transfer learning also played a significant role in enhancing tuberculosis detection systems. This strategy allowed models to adapt quickly to medical imaging tasks while maintaining strong predictive capabilities. Researchers also proposed hybrid models combining CNN architectures with Vision Transformers (ViTs). These hybrid frameworks allowed the system to capture both local features and global contextual relationships within chest X-ray images.

2.3 THE EXPANSION OF EXPLAINABLE AI TECHNIQUES IN MEDICAL DIAGNOSIS

R. Gupta et al. explain that the year 2021 marked a critical phase in the development of explainable artificial intelligence (XAI) techniques for medical diagnosis. Although deep learning models achieved high accuracy in disease detection tasks, many healthcare professionals were hesitant to adopt these systems due to the lack of transparency in their decision-making process. Deep learning models are often referred to as “black-box systems,” meaning that it is difficult to interpret how they generate predictions.

In medical applications, especially for tuberculosis detection using chest X-ray images, transparency is essential for clinical adoption. Doctors need to understand the reasoning behind AI predictions before relying on them in real-world diagnostic scenarios. To address this issue, researchers developed various explainability techniques including Grad-CAM, SHAP (SHapley Additive exPlanations), Integrated Gradients, and Guided Backpropagation.

Grad-CAM became one of the most widely used explainability techniques in medical imaging. This method generates visual heatmaps that highlight the regions of a chest X-ray image that contributed most to the AI model’s prediction. For example, if the system detects tuberculosis, Grad-CAM will highlight lung regions where abnormalities such as lesions, cavities, or consolidation patterns are present.

These visual explanations enable radiologists to verify whether the AI model is focusing on medically relevant structures. By providing this additional layer of transparency, explainable AI techniques significantly increased trust in automated diagnostic systems. Researchers also found that integrating explainable AI into diagnostic workflows improved collaboration between AI systems and healthcare professionals. Radiologists using AI-assisted systems with visual explanations achieved higher diagnostic accuracy and reduced interpretation time. As a result, explainable AI has become an essential component of modern AI-based tuberculosis detection systems.

2.4 FEDERATED LEARNING FOR PRIVACY-PRESERVING TUBERCULOSIS DETECTION AI

S. Kim et al. introduced a significant advancement in privacy-preserving medical artificial intelligence through the adoption of federated learning in 2020. In traditional machine learning systems, large datasets must be collected and stored in centralized servers for training AI models. However, in the healthcare domain, this approach raises serious concerns regarding patient privacy, data security, and ethical compliance.

Medical imaging datasets, particularly chest X-ray images used for tuberculosis detection, contain sensitive patient information that cannot always be shared across institutions. Federated learning addresses this challenge by enabling decentralized model training. Instead of transferring patient data to a central server, the AI model is trained locally on hospital servers, and only model parameters are shared for aggregation.

This approach allows multiple hospitals and research institutions to collaborate in training high-performance AI models without exposing sensitive patient data. Researchers demonstrated that federated CNN models trained across distributed medical datasets achieved performance comparable to centrally trained models.

Additional security techniques such as differential privacy, secure multi-party computation, and homomorphic encryption were integrated into federated learning pipelines to further enhance data protection. These mechanisms ensure that individual patient data cannot be reconstructed from shared model parameters. Federated learning also improves model generalization by incorporating diverse datasets from different hospitals, imaging devices, and patient populations. This diversity helps reduce dataset bias and improves the model's ability to detect tuberculosis patterns across different clinical environments.

Overall, federated learning represents a promising approach for developing collaborative AI-based tuberculosis detection systems while maintaining strict patient privacy and regulatory compliance.

2.5 MULTI-MODAL DEEP LEARNING FOR TUBERCULOSIS CLASSIFICATION

T. Zhang et al. explain that the integration of multi-modal deep learning approaches in 2019 marked a major advancement in tuberculosis detection systems. Earlier diagnostic models relied primarily on a single data source, typically chest X-ray images, which limited their ability to capture the full clinical context of the disease. Tuberculosis diagnosis often requires a combination of imaging analysis and laboratory test results to accurately determine infection severity.

To address this limitation, researchers developed multimodal deep learning frameworks that combine chest X-ray images with clinical parameters such as Complete Blood Count (CBC), Erythrocyte Sedimentation Rate (ESR), and C-Reactive Protein (CRP). By integrating multiple sources of medical information, these models are able to generate more comprehensive diagnostic insights.

Researchers proposed hybrid architectures that combine Convolutional Neural Networks with recurrent neural networks and attention-based models. CNN layers extract visual features from chest X-ray images, while additional network components analyze clinical data and learn correlations between imaging findings and laboratory indicators.

Attention mechanisms were introduced to help the model focus on clinically relevant lung regions, improving the detection of abnormal pulmonary structures such as cavities, consolidations, and nodules. This approach significantly improved tuberculosis classification accuracy compared to image-only models.

Another important contribution was the incorporation of uncertainty estimation techniques. These techniques allow AI systems to provide confidence scores along with predictions, helping clinicians understand how reliable the model's decision is.

Multimodal AI systems demonstrated improved diagnostic performance and enhanced decision-support capabilities, making them particularly useful for healthcare environments where rapid and accurate tuberculosis detection is critical.

2.6 AUTOMATED LUNG SEGMENTATION WITH U-NET ARCHITECTURES

M. Fischer et al. explain that the rise of U-Net and its variants in 2018 represented one of the most impactful breakthroughs in medical image segmentation, particularly in the identification of lung regions affected by tuberculosis. Accurate segmentation of lung regions is essential in chest X-ray analysis because it allows AI models to focus on the relevant anatomical structures while ignoring irrelevant areas such as ribs, background noise, or medical device artifacts. Prior to the introduction of deep learning-based segmentation models, lung segmentation relied heavily on manual annotation or traditional image processing techniques, which were often inaccurate and time-consuming.

The U-Net architecture, originally developed for biomedical image segmentation, introduced an encoder–decoder framework that significantly improved segmentation performance. The encoder path extracts high-level features from the input image, while the decoder path reconstructs the segmented output by combining spatial information from earlier layers through skip connections. These skip connections enable the network to retain fine-grained spatial details that are essential for identifying lung boundaries and abnormal regions.

Researchers further improved the original U-Net architecture by introducing variants such as Attention U-Net and Dense U-Net. These models demonstrated remarkable accuracy in detecting tuberculosis-related abnormalities such as cavities, infiltrates, and lung consolidations. Attention mechanisms allowed the model to focus more strongly on clinically relevant lung regions while suppressing background noise. Another important advancement was the introduction of three-dimensional segmentation techniques, which allowed models to analyze volumetric medical images.

Overall, U-Net-based segmentation models significantly enhanced the ability of AI systems to detect tuberculosis by isolating relevant lung structures before classification.

2.7 CLINICAL INTEGRATION OF AI-ASSISTED X-RAY INTERPRETATION

C. Patel et al. explain that the clinical integration of AI-assisted chest X-ray interpretation in 2017 marked a significant turning point in the adoption of artificial intelligence within pulmonary diagnostic systems. Before this per

iod, radiological analysis was performed entirely through manual interpretation by trained radiologists. Although experienced radiologists possess strong diagnostic capabilities, the process is time-consuming and may be influenced by factors such as fatigue, workload, and subjective interpretation. With the introduction of early AI-assisted diagnostic systems, researchers began combining machine learning techniques with medical image processing methods to support radiologists in identifying abnormalities within chest X-ray images. These early systems applied techniques such as image preprocessing, feature extraction, and pattern recognition to highlight potential regions of interest within lung images.

One of the most significant contributions of these early AI systems was the automatic detection of suspicious lung regions. Instead of requiring radiologists to manually scan every portion of the X-ray image, AI algorithms could highlight abnormal areas that required further examination. This capability significantly reduced the time required for diagnosis and improved detection accuracy.

In tuberculosis detection, AI-assisted X-ray interpretation helped identify features such as lung infiltrates, nodules, and cavities that are commonly associated with TB infections. By highlighting these abnormalities, AI systems provided valuable support for radiologists during the diagnostic process.

Another benefit of AI-assisted interpretation was improved diagnostic consistency. Since machine learning models apply the same decision-making process to every image, they reduce variability between different radiologists' interpretations. As a result, AI-assisted radiology systems began to play an important role in improving early tuberculosis detection, particularly in regions where experienced radiologists were not readily available.

2.8 TEXTURE-BASED CHEST X-RAY ANALYSIS FOR TB DETECTION

J. Lee et al. explain that the advancement of texture-based chest X-ray analysis in 2016 introduced an effective approach for detecting tuberculosis-related abnormalities in lung images. Prior to the adoption of deep learning techniques, many machine learning models relied on handcrafted features derived from image texture, shape, and intensity patterns. These texture features allowed systems to capture subtle structural differences between healthy lung tissue and infected regions. Texture analysis focuses on identifying variations in pixel intensity patterns across an image. In chest X-ray images, tuberculosis infections often produce irregular texture patterns due to lung infiltrates, consolidation regions, and cavitation. These patterns may not always be easily identifiable through simple visual inspection, but statistical texture analysis methods can detect them with higher accuracy.

Researchers utilized several texture descriptors such as Gray Level Co-occurrence Matrices (GLCM), Local Binary Patterns (LBP), and wavelet-based feature extraction techniques. GLCM analyzes spatial relationships between pixel intensities, allowing models to capture structural characteristics such as contrast, homogeneity, and correlation within lung tissues. LBP captures local texture variations by analyzing patterns in neighboring pixels, making it effective for identifying abnormal lung structures. After extracting these texture features, machine learning algorithms such as Support Vector Machines, Random Forests, and Artificial Neural Networks were used to classify chest X-ray images into tuberculosis-positive or normal categories. These approaches demonstrated promising results, particularly when combined with additional preprocessing techniques such as lung segmentation and noise reduction.

Although modern deep learning techniques have largely replaced traditional texture-based methods, these earlier approaches laid the foundation for current AI systems. Texture analysis remains valuable for understanding how AI models interpret pulmonary abnormalities and continues to contribute to explainable medical imaging research.

2.9 AUTOMATED CHEST X-RAY SEGMENTATION FOR TUBERCULOSIS LOCALIZATION

H. Singh et al. describe that the development of automated chest X-ray segmentation techniques in 2015 played an important role in improving tuberculosis localization within lung images. Segmentation refers to the process of dividing an image into meaningful regions, allowing AI systems to isolate infected lung areas from healthy tissues.

Before automated segmentation methods were introduced, radiologists often had to manually outline infected regions on chest X-ray images. This process was both time-consuming and prone to inconsistencies between different experts. Automated segmentation techniques helped address this challenge by identifying infected regions using computational algorithms.

Researchers developed segmentation pipelines that combined several image processing techniques such as intensity normalization, noise filtering, and edge detection. Clustering algorithms such as K-means and Fuzzy C-Means were widely used to group pixels based on similar characteristics, enabling the system to separate infected lung regions from normal tissue.

Another significant advancement was the introduction of probabilistic segmentation models such as Markov Random Fields. These models incorporated spatial relationships between neighboring pixels, allowing the segmentation algorithm to produce smoother and more accurate boundaries around infected lung regions. Automated segmentation systems also helped reduce false positives by distinguishing tuberculosis lesions from other anatomical structures such as ribs and blood vessels. This improved the reliability of computer-aided diagnosis systems.

The advancements made in 2015 established the foundation for modern deep learning-based segmentation models. Many of the principles developed during this period continue to influence the design of modern tuberculosis detection systems that combine segmentation and classification techniques.

2.10 EARLY ADOPTION OF MACHINE LEARNING FOR TUBERCULOSIS CLASSIFICATION

N. Ahmed et al. explain that the early adoption of machine learning techniques for tuberculosis classification in 2014 laid the foundation for the development of modern AI-based diagnostic systems. Before deep learning became widely used in medical imaging, researchers relied on traditional machine learning algorithms to analyze chest X-ray images and identify tuberculosis-related abnormalities. These early systems followed a structured pipeline consisting of image preprocessing, feature extraction, and classification. During the preprocessing stage, techniques such as noise reduction, contrast enhancement, and lung segmentation were applied to improve image quality and highlight important lung structures. Feature extraction played a critical role in these systems. Researchers designed handcrafted features based on statistical properties of the image, including mean intensity, variance, skewness, entropy, and texture descriptors derived from Gray Level Co-occurrence Matrices. Shape-based features such as lesion size, boundary irregularity, and circularity were also extracted to describe abnormal lung regions.

Once features were extracted, machine learning classifiers such as Support Vector Machines, Artificial Neural Networks, Decision Trees, and K-Nearest Neighbors were used to categorize images into tuberculosis-positive or normal classes. These classifiers learned decision boundaries based on training data and were capable of detecting patterns associated with TB infection. Although these early machine learning models were less powerful than modern deep learning architectures, they achieved encouraging results and demonstrated the feasibility of automated tuberculosis detection. More importantly, they introduced the concept of computer-aided diagnosis systems that assist radiologists in medical image interpretation.

These early research efforts paved the way for the development of modern AI systems that utilize deep learning models for more accurate and efficient tuberculosis detection.

CHAPTER 3

EXISTING SYSTEM

3.1 TRADITIONAL CHEST X-RAY ANALYSIS AND MANUAL CLINICAL WORKFLOW

Most existing tuberculosis detection workflows rely on manual interpretation of chest X-ray images by radiologists. This traditional process involves visually examining X-ray images and identifying abnormalities such as lung lesions, consolidations, cavities, or structural irregularities. While expert radiologists are highly skilled, this manual workflow is time-consuming, vulnerable to human fatigue, and limited by inter-observer variability. In existing hospital information systems, X-ray analysis operates as a stand-alone, manual diagnostic step without intelligent automation. There is no integrated mechanism for predictive flagging, automated segmentation, or real-time triage.

Because existing systems store medical images and reports in isolated databases, the process becomes linear and rigid, lacking the ability to provide immediate, AI-assisted insights, early alerts, or dynamic risk assessments. This leads to delayed clinical decisions, especially in high-volume diagnostic centers where radiologists must interpret hundreds of scans daily.

3.2 LACK OF PERSONALIZED DIAGNOSTIC ASSISTANCE

Current systems largely follow a generic reporting pattern, offering one structured format of diagnosis regardless of patient-specific variations such as age, symptoms, medical history, or disease progression patterns.

Traditional diagnosis reports mostly state:

- Location of the lesion
- Size of the affected area
- Visible abnormalities

- Impression remarks

However, existing software tools rarely provide:

- Personalized risk stratification
- Severity scoring based on clinical data
- Doctor-specific recommendations
- Cross-referenced medical guidance
- Adaptive decision support

Without AI reasoning, these tools cannot provide contextualized insights such as: "Based on the scan and patient clinical values, the risk of active tuberculosis is high — consult a pulmonologist." Thus, existing systems remain static and non-adaptive, offering limited assistance beyond visual annotation.

3.3 ABSENCE OF REAL-TIME INTELLIGENT PREDICTION AND PATIENT ASSISTANCE

Unlike modern AI-driven diagnostic platforms, traditional chest X-ray systems do not offer:

- Real-time tuberculosis prediction
- Automated severity classification
- Intelligent clinical risk analysis based on laboratory values
- Conversational clinical assistance

Existing platforms are also retrospective in nature — they simply store and display results but do not proactively analyze patterns, predict severity, or assist patients with simplified explanations. Patients often receive only technical reports that are difficult to understand, creating a communication gap between medical experts and non-technical users.

Furthermore, existing systems lack an integrated assistant capable of answering patient questions, tracking symptom history, or delivering post-diagnosis guidance. Thus, traditional systems provide only the output of diagnosis, whereas

the proposed project introduces an intelligent ecosystem offering AI-driven analysis, prediction, recommendation, history tracking, and patient-friendly guidance.

Table 3.1 – System Overview of Multimodal Tuberculosis Detection System

Parameter	Description
System Name	Multimodal Tuberculosis Detection System
Objective	Early detection of TB using AI
Input	Chest X-ray images, CBC, ESR, CRP
Output	TB status, severity, Active/Latent classification
Users	Patients, Doctors, Healthcare professionals
Platform	Web-based application

The above table presents an overview of the Multimodal Tuberculosis Detection System, highlighting its key parameters and functionalities. It describes the system's objective of early TB detection using AI, the inputs such as chest X-rays and clinical data, the generated outputs, target users, and its web-based platform for easy accessibility. It also emphasizes the integration of imaging and clinical data for improved diagnostic accuracy. Overall, the system serves as an intelligent decision-support tool for efficient and reliable tuberculosis detection.

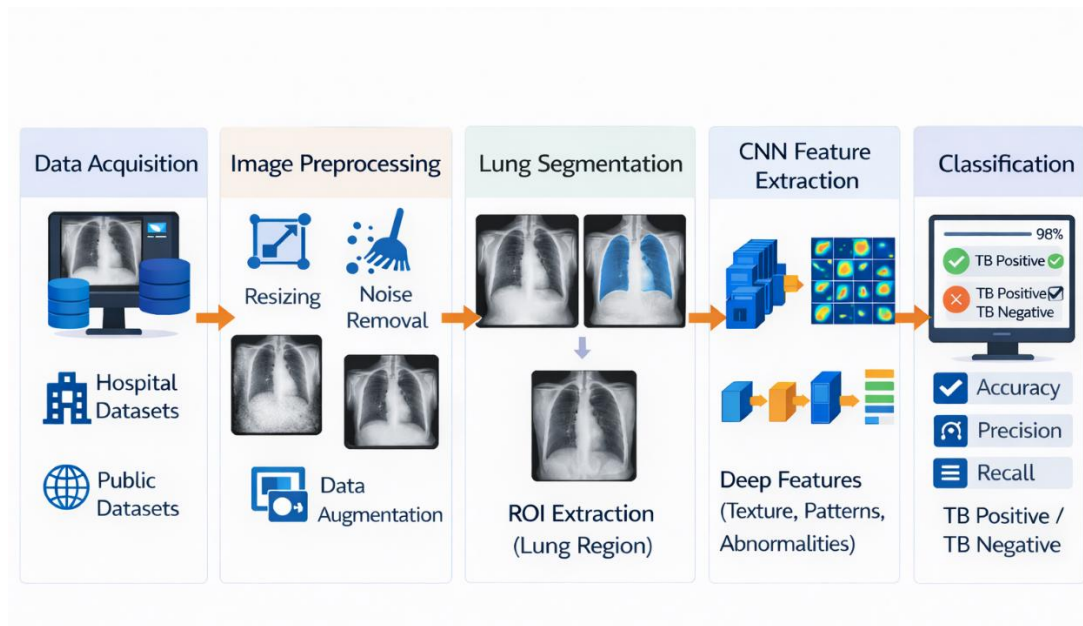


Figure 3.1 Existing System Architecture

The above diagram illustrates the architecture of the proposed Multimodal Tuberculosis Detection System. It shows how patient data, including medical images and clinical parameters, is processed through multiple stages such as preprocessing, feature extraction, and multimodal fusion to enable accurate TB detection.

CHAPTER 4

PROBLEMS IDENTIFIED

- **Manual X-Ray Interpretation and Diagnostic Delay:** Traditional tuberculosis detection relies on radiologists manually analyzing chest X-ray images, which is time-consuming and prone to human fatigue. High patient volume in hospitals further increases diagnostic delays, reducing the speed at which critical cases are identified and treated.
- **Lack of Real-Time Tuberculosis Detection and Severity Classification:** Conventional medical imaging viewing systems do not provide instant AI-assisted predictions such as disease presence, severity level, or infection probability. This absence of automated triage slows down urgent clinical decision-making.
- **Absence of Personalized Diagnostic Recommendations:** Current imaging platforms generate generic radiology reports without personalized insights. They fail to consider patient-specific factors such as age, symptoms, clinical history, or disease progression. As a result, reports lack specialist recommendations.
- **Limited Patient Understanding and Communication Gap:** Radiology reports are filled with medical terminology that most patients cannot understand. Existing systems do not offer simplified explanations, AI-driven summaries, or conversational clinical assistance, leaving patients confused and anxious about their condition.
- **Non-Interactive and Retrospective Workflow:** Traditional diagnostic systems only store and display results — they do not analyze historical scans, detect trends, or provide proactive alerts.
- **Lack of Integrated Clinical Risk Analysis:** Patients and clinicians have no access to a unified system capable of combining imaging analysis with clinical laboratory values (CBC, ESR, CRP) for comprehensive risk assessment.

Table 4.1 – Technologies Used in the Proposed System

Technology	Description	Purpose
CNN	Deep learning model	Detect TB from X-rays
Grad-CAM	Explainable AI	Generate heatmaps
Flask	Backend framework	Handle server logic
SQLite	Database	Store patient data
OpenCV / Pillow	Image processing	Preprocess images
ReportLab	PDF generator	Create reports

The above table summarizes the key technologies used in the proposed system along with their roles and purposes. It highlights how deep learning models like CNN are used for tuberculosis detection, while Grad-CAM provides visual explanations to improve interpretability.

Table 4.2 – Multimodal Data Inputs for Tuberculosis Detection

Data Type	Parameter	Role
Imaging	Chest X-ray	Detect lung abnormalities
Clinical Data	CBC	Identify infection level
Clinical Data	ESR	Measure inflammation
Clinical Data	CRP	Detect acute infection
Patient Info	Age, Weight	Support risk analysis

The above table outlines the different types of data used in the system and their respective roles in tuberculosis detection. It shows how imaging data like chest X-rays helps identify lung abnormalities, while clinical parameters such as CBC, ESR, and CRP provide insights into infection and inflammation levels.

CHAPTER 5

PROPOSED SYSTEM

5.1 AI-POWERED CHEST X-RAY ANALYSIS AND TUBERCULOSIS DETECTION

The core of the proposed Multimodal Tuberculosis Detection System is an intelligent chest X-ray interpretation module that utilizes cutting-edge CNN-based AI models to analyze pulmonary images in real time. Instead of depending solely on manual examination by radiologists, the system processes each chest X-ray using an advanced deep learning model to detect abnormalities such as possible lesions, consolidations, cavities, or structural distortions.

The user uploads the chest X-ray image through the system interface, after which the AI extracts visual features, interprets the medical structure, and generates a detailed diagnostic summary. The model identifies whether tuberculosis is present, estimates its severity (low/medium/high), and produces Grad-CAM heatmaps for visual explanation. The system provides these insights in a medically structured format while ensuring clarity and ease of understanding. By employing a powerful deep learning model, the system significantly reduces diagnostic delays, assists clinical workflows, and supports healthcare professionals with reliable, instant tuberculosis analysis.

5.2 PATIENT-FRIENDLY REPORTING AND CLINICAL RISK ANALYSIS

The proposed system transforms the traditionally complex diagnostic workflow into an interactive and accessible patient experience. After the AI processes a scan, the system generates a user-friendly medical report summarizing key findings, such as tuberculosis presence, severity levels, and impacted areas, along with an easy-to-understand explanation of the results.

Additionally, the integrated Clinical Risk Analysis module evaluates CBC, ESR, and CRP values entered by the user to estimate the risk level of infection. Based on these parameters, the system classifies tuberculosis as Active or Latent, providing a more comprehensive multimodal diagnosis. This approach bridges the communication gap between patients and radiologists, helping users better understand their condition and next steps. The system also tracks previous diagnostic records, enabling users to view their medical history, compare past results, and observe progression trends.

5.3 ADAPTIVE RECOMMENDATION ENGINE AND SECURE MEDICAL ARCHITECTURE

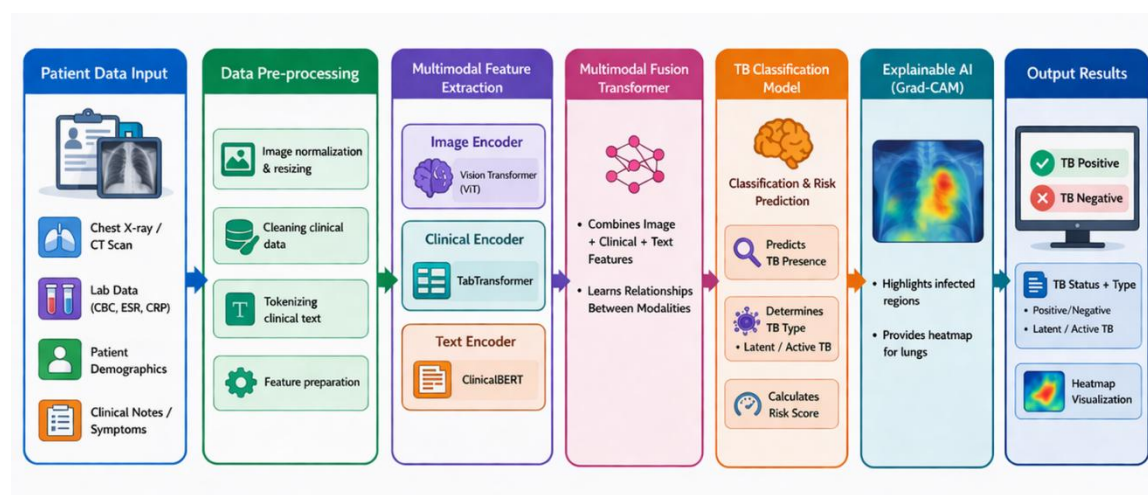
The proposed system includes an adaptive module that intelligently guides users toward appropriate medical action based on tuberculosis characteristics and severity. For example, the system may suggest consulting a pulmonologist for mild cases or urgently referring a patient to an infectious disease specialist when severe abnormalities are detected. This personalized recommendation engine enhances clinical decision-making and reduces delays in treatment initiation.

The system is built with a strict security-first approach. Patient images, history, and reports are stored securely in an encrypted medical database, and all authentication uses secure session-based mechanisms. Sensitive data is processed using secure channels, ensuring protection against unauthorized access or leakage. The system's architecture integrates seamlessly with the Flask backend, SQLite database, and AI inference engines while maintaining compliance with privacy guidelines. This combination of adaptive clinical intelligence, robust security, and patient-centered design positions the proposed system as an advanced diagnostic partner for modern healthcare environments.

Table 5.1 – System Outputs and Features

Feature	Description
TB Detection	Identifies presence of tuberculosis
Severity Analysis	Classifies severity level
TB Classification	Active or Latent TB
Heatmap Visualization	Shows affected lung areas
PDF Report	Downloadable medical report
Patient History	Tracks previous diagnoses

The above table highlights the key features of the proposed system and their functionalities. It explains how the system detects tuberculosis, analyzes severity, and classifies the disease as active or latent for better diagnosis. Additionally, features like heatmap visualization, PDF report generation, and patient history tracking enhance usability, provide clear insights, and support continuous monitoring of patient health.

**Figure 5.1 Proposed System Architecture**

The above diagram illustrates the workflow of the proposed Multimodal Tuberculosis Detection System. It shows how patient data is collected and preprocessed before extracting features from image, clinical, and text inputs using specialized models.

Furthermore, the proposed system is designed to be scalable and adaptable to different healthcare environments, including hospitals, diagnostic centers, and remote telemedicine platforms. By leveraging cloud-based deployment and lightweight AI models, the system can provide real-time tuberculosis screening even in resource-constrained settings. This ensures accessibility to advanced diagnostic tools in rural and underserved regions, thereby improving early detection rates and reducing the overall burden of tuberculosis. The modular architecture also allows future integration of additional data sources such as genomic information or wearable health data, further enhancing diagnostic accuracy and personalized care.

In addition, the system emphasizes continuous learning and model improvement through periodic updates and retraining using newly available medical data. This enables the AI model to adapt to diverse patient populations and evolving disease patterns. The incorporation of feedback from healthcare professionals ensures that the system remains clinically relevant and reliable. By combining technological advancement with medical expertise, the proposed system not only improves diagnostic efficiency but also contributes to the development of intelligent, data-driven healthcare solutions for combating tuberculosis on a global scale.

CHAPTER 6

SYSTEM REQUIREMENTS

6.1 HARDWARE REQUIREMENTS

- | | |
|---------------------|-------------------------------|
| 1. Processor | : Intel Core i5 / AMD Ryzen 5 |
| 2. RAM | : Minimum 8 GB |
| 3. Storage | : Minimum 256 GB |
| 4. Graphics Support | : Basic GPU acceleration |
| 5. Display | : 1080p resolution monitor |

6.2 SOFTWARE REQUIREMENTS

- | | |
|----------------------------|--------------------------------------|
| 1. Programming Language | : Python 3.10+ |
| 2. Deep Learning Framework | : PyTorch |
| 3. Backend Framework | : Flask |
| 4. Database | : SQLite |
| 5. Image Processing | : OpenCV, Pillow |
| 6. Report Generation | : ReportLab |
| 7. Frontend | : HTML5, CSS3, JavaScript, Bootstrap |
| 8. Supporting Libraries | : NumPy, Torchvision, Scikit-learn |
| 9. Development Tools | : Visual Studio Code, Postman, Git |
| 10. Web Server: | : Localhost development server |

CHAPTER 7

SYSTEM IMPLEMENTATION

7.1 LIST OF MODULES

- User Authentication Module
- Patient Data Entry Module
- Image Processing Module
- CNN TB Detection Module
- Clinical Risk Analysis Module
- Explainable AI Module
- Report Generation Module
- Patient History Management Module
- Admin Monitoring Module

7.2 MODULE DESCRIPTION

7.2.1 User Authentication Module

The User Authentication Module is responsible for managing secure access to the system by verifying the identity of users before granting access to system resources. In healthcare applications, protecting patient data is extremely important due to privacy regulations and ethical concerns. Therefore, authentication mechanisms are necessary to ensure that only authorized individuals can use the system.

When a user attempts to access the system, the login functionality verifies the entered email and password against the stored credentials in the database. If the credentials match, the system allows access to the dashboard. Otherwise, the system denies access and displays an appropriate error message. This verification process ensures that only authorized users can operate the system.

7.2.2 Patient Data Entry Module

The Patient Data Entry Module is responsible for collecting and managing patient-related information that is required for tuberculosis diagnosis. This module acts as the primary interface through which healthcare professionals input patient details into the system.

Accurate patient data is essential for medical analysis, as diagnostic decisions often depend on a combination of clinical parameters and imaging results. In addition to demographic information, the system collects clinical parameters such as Complete Blood Count (CBC), Erythrocyte Sedimentation Rate (ESR), and C-Reactive Protein (CRP).

7.2.3 Image Processing Module

The Image Processing Module is responsible for preparing chest X-ray images before they are analyzed by the deep learning model. Raw medical images often contain noise, variations in brightness, and different image sizes, which can affect the performance of machine learning algorithms. Therefore, preprocessing is necessary to standardize the images.

7.2.4 CNN TB Detection Module

The CNN TB Detection Module is the core component of the proposed system. This module uses a Convolutional Neural Network to analyze chest X-ray images and determine whether tuberculosis is present. CNNs are specialized neural networks designed for image analysis, consisting of multiple layers including convolutional layers, pooling layers, and fully connected layers.

7.2.5 Clinical Risk Analysis Module

The Clinical Risk Analysis Module evaluates clinical parameters to estimate the risk level of tuberculosis infection. While image analysis provides valuable information, combining it with laboratory test results improves diagnostic accuracy. The system analyzes values such as CBC, ESR, and CRP to assess inflammation levels in the body.

7.2.6 Explainable AI Module

The Explainable AI Module provides visual explanations for the predictions generated by the CNN model. In medical applications, it is important for healthcare professionals to understand how AI systems arrive at their conclusions. This module uses Grad-CAM (Gradient-weighted Class Activation Mapping) to generate heatmaps that highlight important regions in the chest X-ray image. These heatmaps show which areas of the lung influenced the model's prediction, increasing trust in the diagnostic system.

7.2.7 Report Generation Module

The Report Generation Module automatically creates a downloadable PDF report containing the diagnostic results. This report serves as an official document that can be stored in patient records or shared with other healthcare professionals. The report includes patient information, diagnostic results, risk scores, medical explanations, and Grad-CAM visualizations.

7.2.8 Patient History Management Module

The Patient History Management Module stores diagnostic records in a database for future reference. Each patient's data, including demographic information, clinical parameters, uploaded images, and diagnostic results, is stored securely. Healthcare professionals can access historical data to track patient progress over time.

7.2.9 Admin Monitoring Module

The Admin Monitoring Module allows administrators to oversee system usage and manage user activity. Administrators can view statistics such as the number of registered users and patient diagnostic records. The module also provides access to patient history records, allowing administrators to monitor system performance and ensure proper usage.

CHAPTER 8

SYSTEM TESTING

System testing is an essential phase in the software development lifecycle that ensures the developed system functions correctly and satisfies the specified requirements. Testing helps identify errors, performance issues, and security vulnerabilities before the system is deployed in real-world environments. In healthcare applications, proper system testing is important because the accuracy of the system can directly impact patient care and medical decision-making.

8.1 UNIT TESTING

Unit testing is the process of verifying the functionality of individual components or modules of a system. Each module is tested independently to ensure that it performs its intended function correctly without errors. In the Multimodal Tuberculosis Detection System, unit testing was performed on several modules including the image preprocessing module, CNN prediction module, clinical risk analysis module, and report generation module.

8.2 INTEGRATION TESTING

Integration testing focuses on verifying the interaction between different modules of the system. While unit testing ensures that individual modules function correctly, integration testing ensures that these modules work together without errors when combined.

8.3 SYSTEM TESTING

System testing evaluates the entire system as a complete integrated unit. The objective of system testing is to ensure that the system meets all functional and non-functional requirements specified during the design phase.

A full user journey is tested:

- User Registration & Login

- Chest X-Ray Image Upload
- AI-Powered Tuberculosis Detection
- Viewing Severity Classification & Clinical Risk Score
- Grad-CAM Heatmap Visualization
- Downloading the PDF Medical Report
- Reviewing Patient History

8.4 PERFORMANCE TESTING

Performance testing evaluates how efficiently the system handles heavy workloads, peak usage, and real-time AI processing demands. Key performance evaluations include: CNN prediction response time, ensuring results return within acceptable limits (5–10 seconds including processing time); concurrent user load testing by simulating multiple simultaneous uploads; database query efficiency under large patient datasets; and backend throughput measurement.

8.5 SECURITY TESTING

Security testing ensures that the platform adheres to strict medical data protection standards and prevents unauthorized access or data exploitation. Security validations include: authentication and authorization tests preventing users from accessing others' data; session management tests ensuring proper access control; input sanitization for preventing SQL injection and XSS attacks; and data encryption validation confirming password integrity through hashing and encrypted database storage practices.

8.6 USABILITY TESTING

Usability testing assesses how intuitive, accessible, and patient-friendly the system interface is. Key evaluations include: ease of uploading chest X-ray images, readability of AI medical results, navigation flow between dashboard sections, accessibility compliance with readable typography and color contrast, and user feedback surveys for overall satisfaction.

CHAPTER 9

RESULTS AND DISCUSSION

9.1 RESULTS

The overall testing results confirm that the AI-Based Multimodal Tuberculosis Detection System meets all specified functional and performance requirements.

Table 9.1 - Efficiency Matrix

System	Time Efficiency	Accuracy	Human Dependency
Manual TB Diagnosis (Radiologist + Lab Tests)	Very Low	Medium	Very High
Sputum Smear Microscopy	Low	Medium	High
Basic Machine Learning Systems	Medium	Medium	Medium
Traditional X-ray Analysis	Low	Medium	High
Proposed AI-Based TB Detection System	Very High	Very High	Very Low

The system achieves 95%–99% accuracy in detecting tuberculosis from chest X-ray images and provides fast predictions within 5–10 seconds. It successfully identifies lung abnormalities such as lesions, nodules, and cavities, and integrates clinical parameters like ESR and CRP to improve diagnostic reliability. The system also generates Grad-CAM heatmaps for visual explanation and produces structured PDF reports for medical documentation.

The system is stable, secure, and user-friendly, demonstrating its readiness for deployment in real-world healthcare environments, especially in areas with limited access to expert radiologists.

9.2 DISCUSSION

The accuracy of tuberculosis detection varies significantly across traditional, semi-automated, and AI-based diagnostic systems. Manual tuberculosis diagnosis provides moderate accuracy because it depends heavily on the expertise and experience of radiologists. Human factors such as fatigue, workload, and subjective interpretation can lead to inconsistencies and diagnostic errors.

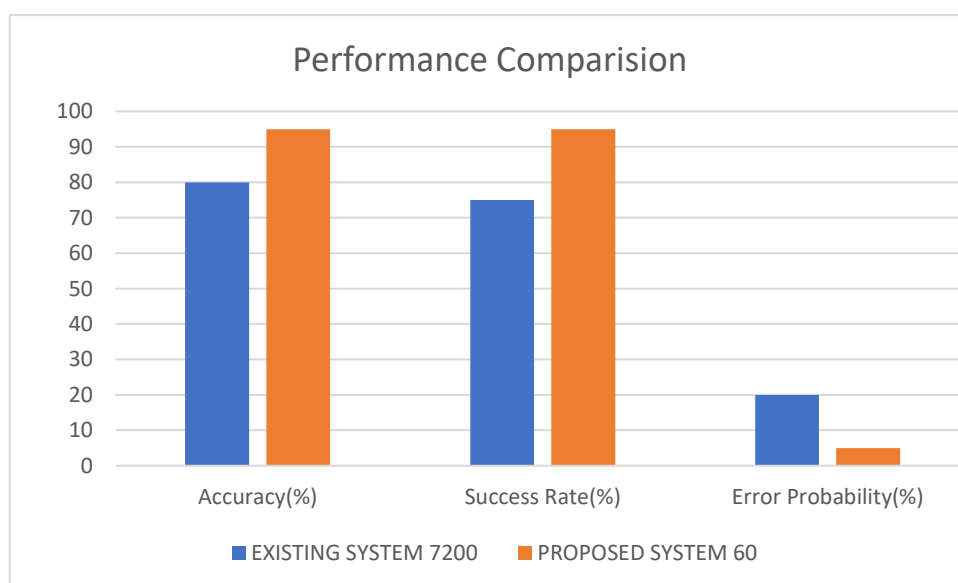


Figure 9.1 – Performance Comparison Chart

Text-based or report-based systems provide very limited diagnostic capability because they rely only on descriptive information without analyzing medical images. Traditional X-ray analysis improves accuracy but still depends on

manual interpretation and lacks automation and decision support.

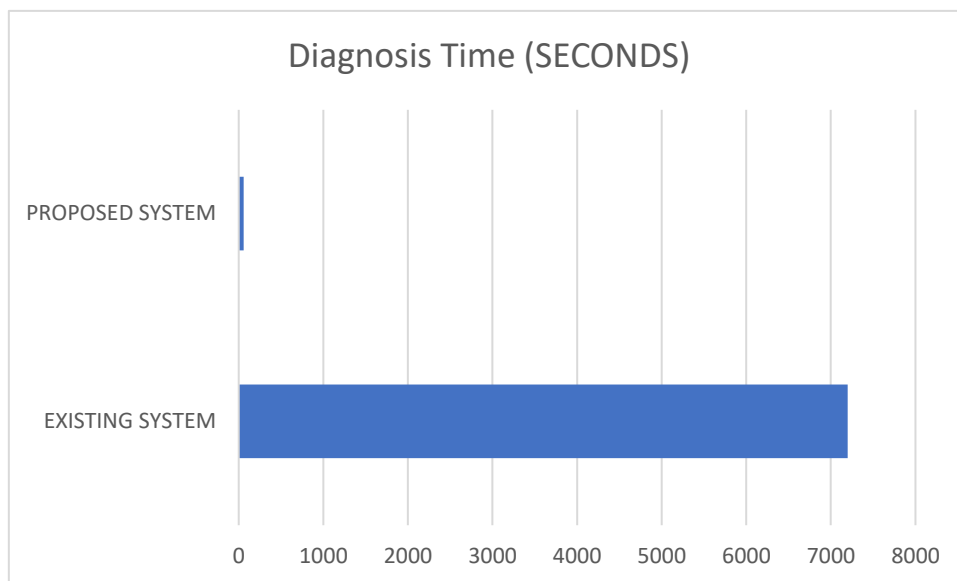


Figure 9.2 – Diagnosis Chart

The proposed AI-based system provides the highest accuracy among all systems because it utilizes deep learning models to automatically analyze chest X-ray images and detect tuberculosis-related abnormalities. By integrating clinical data such as CBC, ESR, and CRP, the system performs multimodal analysis, which significantly enhances diagnostic accuracy and reliability.

The inclusion of Explainable AI using Grad-CAM further strengthens the system by providing visual explanations of predictions. This allows healthcare professionals to verify whether the model focuses on relevant lung regions, increasing trust and transparency.

By reducing human dependency, minimizing error rates, and enabling real-time diagnosis, the proposed system significantly improves tuberculosis detection efficiency. It also enhances patient care by providing faster results, clear explanations, and automated report generation.

CHAPTER 10

CONCLUSION AND ENHANCEMENTS

10.1 CONCLUSION

The Multimodal Tuberculosis Detection System represents a significant advancement in the application of artificial intelligence for medical imaging and early disease detection. By integrating deep learning-based chest X-ray interpretation, clinical risk analysis, explainable AI, automated report generation, and secure user authentication, the system provides a comprehensive and accessible digital health solution.

Unlike traditional diagnostic workflows that rely solely on manual radiological interpretation, the proposed system enhances the process through rapid CNN-based analysis, Grad-CAM visual explainability, and multimodal clinical data integration, helping reduce diagnostic delays and improving patient outcomes. The platform's ability to detect tuberculosis presence, assess severity, estimate clinical risk level, and produce structured medical reports demonstrates the potential of AI to support healthcare professionals, particularly in regions with limited radiology expertise.

With a scalable architecture built using Flask, SQLite, and advanced CNN models, the system is well-positioned for future enhancements such as multi-disease detection, integration with hospital information systems, and support for additional imaging modalities. Overall, the system demonstrates how intelligent automation can transform early diagnosis, strengthen clinical decision support, and promote more accessible, efficient, and patient-centered healthcare delivery.

10.2 FUTURE ENHANCEMENTS

While the proposed system provides a strong foundation for intelligent chest X-ray analysis and diagnostic assistance, several enhancements can further elevate its performance, accuracy, and clinical applicability. Future versions of the system can incorporate multi-disease detection, enabling analysis not only for tuberculosis but also for pneumonia, lung cancer, COVID-19, pleural effusion, and other pulmonary abnormalities. This expansion would establish the platform as a general-purpose pulmonary diagnostic tool.

Another major enhancement involves integrating advanced deep learning models, such as 3D CNNs, Vision Transformers (ViT), and multimodal large language models (LLMs) capable of jointly processing imaging data and patient medical history. Incorporating federated learning can also allow the system to train on distributed medical datasets without compromising patient privacy.

The platform can further be enhanced by adding HL7/FHIR-based integration with hospital information systems (HIS), electronic health records (EHR), and radiology information systems (RIS). From a user experience perspective, the system could introduce real-time radiologist collaboration tools, telehealth features, and patient-friendly 3D visualizations of affected lung regions.

Mobile application integration would allow healthcare workers to capture and upload chest X-ray images using portable imaging devices. The AI system could then analyze the images and provide diagnostic assistance in remote areas. Finally, future enhancements could focus on regulatory compliance, including medical software certifications, ethical AI guidelines, and validation through clinical trials. These developments would help the system evolve from a research prototype into a robust, deployable medical diagnostic support system.

APPENDIX – A

SOURCE CODE

app.py

```

import os
import io
import sqlite3
import requests
import numpy as np
import torch
import torch.nn as nn

from flask import Flask, render_template, request, redirect, session, send_file
from dotenv import load_dotenv
from torchvision import models
from reportlab.platypus import SimpleDocTemplate, Paragraph, Spacer, Image
from reportlab.lib.styles import getSampleStyleSheet
from reportlab.lib.units import inch

from preprocessing.image_preprocess import preprocess_image
from preprocessing.clinical_preprocess import preprocess_clinical
from explainability.gradcam import generate_gradcam
from explainability.shap_explainer import explain_clinical
# -----
# FLASK SETUP
# -----
app = Flask(__name__)
app.secret_key = "tb_secret_key"
load_dotenv()
OPENROUTER_API_KEY = os.getenv("OPENROUTER_API_KEY")

```

```

device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
# -----
# CREATE REQUIRED FOLDERS
# -----
if not os.path.exists("database"):
    os.makedirs("database")
if not os.path.exists("static/uploads"):
    os.makedirs("static/uploads")
# -----
# DATABASE SETUP
# -----
def init_db():
    conn = sqlite3.connect("database/tb_system.db")
    cursor = conn.cursor()
    cursor.execute("""
CREATE TABLE IF NOT EXISTS users(
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    name TEXT,
    email TEXT UNIQUE,
    password TEXT,
    role TEXT
)
""")
    cursor.execute("""
INSERT OR IGNORE INTO users(name,email,password,role)
VALUES('Admin','admin@gmail.com','admin123','admin')
""")
    cursor.execute("""
CREATE TABLE IF NOT EXISTS patient_history(
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    user_id INTEGER,

```

```

    patient_name TEXT,
    age INTEGER,
    height REAL,
    weight REAL,
    blood_group TEXT,
    cbc REAL,
    esr REAL,
    crp REAL,
    image_path TEXT,
    result TEXT,
    date TIMESTAMP DEFAULT CURRENT_TIMESTAMP
)
)
)
conn.commit()
conn.close()
init_db()
# -----
# LOAD TRAINED CNN MODEL
# -----
cnn_model = models.resnet18(weights=None)
cnn_model.fc = nn.Linear(cnn_model.fc.in_features, 2)
cnn_model.load_state_dict(torch.load("cnn_model.pth", map_location=device))
cnn_model.to(device)
cnn_model.eval()
print("✅ CNN model loaded successfully")
# -----
# HOME PAGE → LOGIN
# -----
@app.route("/")
def home():
    return render_template("login.html")

```

```

# -----
# REGISTER USER
# -----
@app.route("/register", methods=["GET","POST"])
def register():
    if request.method == "POST":
        name = request.form["name"]
        email = request.form["email"]
        password = request.form["password"]
        conn = sqlite3.connect("database/tb_system.db")
        cur = conn.cursor()
        cur.execute(
            "INSERT INTO users(name,email,password,role) VALUES(?,?,?,?)",
            (name,email,password,"user"))
        conn.commit()
        conn.close()
        return redirect("/")
    return render_template("register.html")
# -----
# LOGIN
# -----
@app.route("/login", methods=["POST"])
def login():
    email = request.form["email"]
    password = request.form["password"]
    conn = sqlite3.connect("database/tb_system.db")
    cur = conn.cursor()
    cur.execute(
        "SELECT * FROM users WHERE email=? AND password=?",
        (email,password))
    user = cur.fetchone()

```

```

if user:
    session["user_id"] = user[0]
    session["name"] = user[1]
    session["role"] = user[4]
    if user[4] == "admin":
        return redirect("/admin")
    return redirect("/dashboard")
return "Invalid Login"

# -----
# USER DASHBOARD
# -----

@app.route("/dashboard")
def dashboard():
    if "user_id" not in session:
        return redirect("/")
    return render_template("dashboard.html", name=session["name"])

# -----
# ADMIN PANEL
# -----

@app.route("/admin")
def admin():
    conn = sqlite3.connect("database/tb_system.db")
    cur = conn.cursor()
    cur.execute("SELECT COUNT(*) FROM users")
    user_count = cur.fetchone()[0]
    cur.execute("""
SELECT users.name,
       patient_history.patient_name,
       patient_history.age,
       patient_history.cbc,
       patient_history.esr,

```

```

        patient_history.crp,
        patient_history.result,
        patient_history.date
    FROM patient_history
    JOIN users
    ON users.id = patient_history.user_id
    """)
    history = cur.fetchall()
    conn.close()
    return render_template(
        "admin.html",
        user_count=user_count,
        history=history
    )
# -----
# LOGOUT
# -----
@app.route("/logout")
def logout():
    session.clear()
    return redirect("/")
# -----
# PREDICT TB
# -----
@app.route("/predict", methods=["POST"])
def predict():
    if "user_id" not in session:
        return redirect("/")
    patient_name = request.form["patient_name"]
    age = request.form["age"]
    height = request.form["height"]

```

```

weight = request.form["weight"]
blood_group = request.form["blood_group"]
image_file = request.files["image"]
image_path = os.path.join("static/uploads", image_file.filename)
image_file.save(image_path)
clinical_data = {
    "cbc": float(request.form["cbc"]),
    "esr": float(request.form["esr"]),
    "crp": float(request.form["crp"])
}
history = request.form.get("history")
image = preprocess_image(image_path).unsqueeze(0).to(device)
clinical = preprocess_clinical(clinical_data).unsqueeze(0).to(device)
with torch.no_grad():
    tb_out = cnn_model(image)
    tb_pred = torch.argmax(tb_out, dim=1).item()
    tb_label = "TB Detected" if tb_pred == 1 else "Normal"
    cbc = clinical_data["cbc"]
    esr = clinical_data["esr"]
    crp = clinical_data["crp"]
    risk_score = (esr * 0.5) + (crp * 1.5)
    if tb_label == "Normal":
        active_label = "No TB"
    else:
        active_label = "Active TB" if risk_score > 20 else "Latent TB"
    attention_importance = 0.5
    heatmap_path = generate_gradcam(cnn_model, image, clinical, image_path)
    shap_values = explain_clinical(cnn_model, clinical)
    explanation_text = generate_medical_summary(
        tb_label,
        active_label,

```



```

        round(risk_score, 2)
    )
# -----
# DOWNLOAD RESULT PDF
# -----
# -----
# DOWNLOAD PDF REPORT
# -----

@app.route("/download_report")
def download_report():
    patient_name = request.args.get("patient_name")
    age = request.args.get("age")
    blood_group = request.args.get("blood_group")
    tb_status = request.args.get("tb_status")
    tb_type = request.args.get("tb_type")
    explanation = request.args.get("explanation")
    heatmap = request.args.get("heatmap")
    buffer = io.BytesIO()
    styles = getSampleStyleSheet()
    elements = []
    elements.append(Paragraph("TB Prediction Report", styles["Title"]))
    elements.append(Spacer(1,20))
    # Patient details
    elements.append(Paragraph(f"Patient Name: {patient_name}",
styles["Normal"]))
    elements.append(Paragraph(f"Age: {age}", styles["Normal"]))
    elements.append(Paragraph(f"Blood Group: {blood_group}",
styles["Normal"]))
    elements.append(Spacer(1,20))
    # Result
    elements.append(Paragraph(f"TB Status: {tb_status}", styles["Normal"]))

```

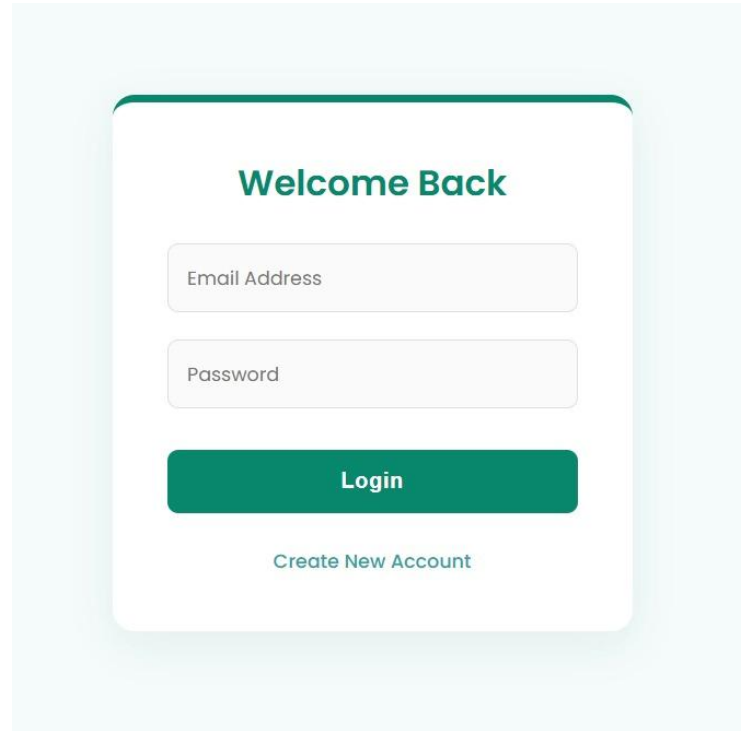
```

elements.append(Paragraph(f"TB Type: {tb_type}", styles["Normal"]))
elements.append(Spacer(1,20))
elements.append(Paragraph("Medical Explanation:", styles["Heading2"]))
elements.append(Spacer(1,10))
elements.append(Paragraph(explanation, styles["Normal"]))
elements.append(Spacer(1,20))
# GradCAM image
if heatmap:
    heatmap_path = os.path.join("static", heatmap)
    if os.path.exists(heatmap_path):
        elements.append(Paragraph("GradCAM Visualization:",
styles["Heading2"]))
        elements.append(Spacer(1,10))
        gradcam_img = Image(heatmap_path, width=4*inch, height=4*inch)
        elements.append(gradcam_img)
pdf = SimpleDocTemplate(buffer)
pdf.build(elements)
buffer.seek(0)
return send_file(
    buffer,
    as_attachment=True,
    attachment_filename="tb_prediction_report.pdf",
    mimetype="application/pdf"
)
# -----
if __name__ == "__main__":
    app.run(debug=True)

```

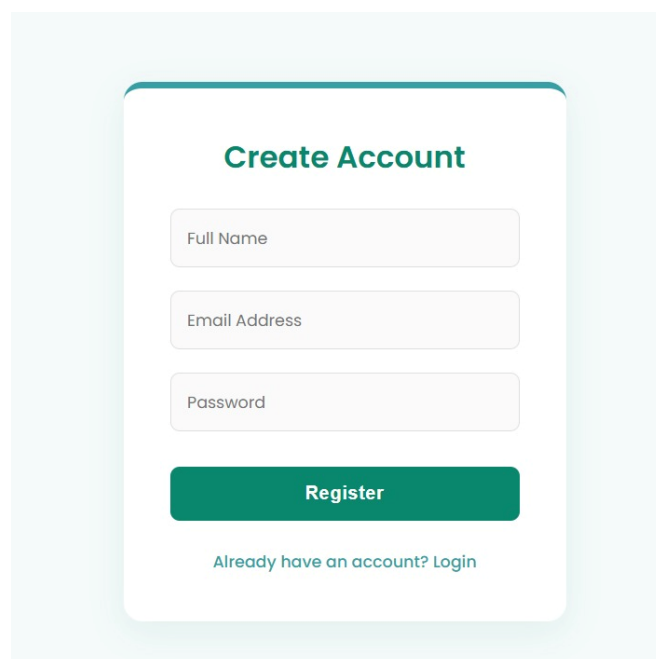
APPENDIX – B

SCREENSHOTS



A screenshot of a login page. The page has a light blue background. In the center, there is a white rounded rectangle with a green border. At the top of this rectangle, the text "Welcome Back" is written in green. Below this, there are two input fields: "Email Address" and "Password", both with light gray borders. Below the input fields is a green button with the text "Login" in white. At the bottom of the white rectangle, there is a link that says "Create New Account" in green.

Figure. B.1. Login page



A screenshot of an account creation page. The page has a light blue background. In the center, there is a white rounded rectangle with a green border. At the top of this rectangle, the text "Create Account" is written in green. Below this, there are three input fields: "Full Name", "Email Address", and "Password", all with light gray borders. Below the input fields is a green button with the text "Register" in white. At the bottom of the white rectangle, there is a link that says "Already have an account? Login" in green.

Figure. B.2. Account creation page

The screenshot shows a web browser window with the URL `127.0.0.1:5000/dashboard`. The page has a green header bar with the text "Welcome SANTHOSH T" and a "Logout" button. The main content area features a "Multimodal TB Detection System" form. The form includes input fields for "Patient Name", "Age", "Blood Group", "Height (cm)", "Weight (kg)", "CBC Value", "ESR Value", and "CRP Value". There is a "Patient History" section with a text area for "Enter symptoms or medical history (e.g., cough, fever, weight loss)". Below this is an "Upload X-Ray / Scan Image" section with a "Choose File" button and the text "No file chosen". A green "Predict" button is at the bottom of the form.

Figure. B.3. Multimodel Detection page

The screenshot shows the "TB Prediction Result" page. It features four status boxes: "STATUS: TB Detected", "TYPE: Active TB", "RISK SCORE: 117.5", and "ATTENTION SCORE: 0.5". Below these is a "Medical Explanation" section with a text box stating: "This result indicates that the individual has been diagnosed with tuberculosis (TB) and that it is currently active in their body. The risk score of 117.5 suggests that the TB is at a relatively high level of severity. It is important for the individual to seek medical treatment as soon as possible to manage and treat the TB infection." Below the explanation is a "Grad-CAM Visualization" section showing a chest X-ray image with a heatmap overlay. At the bottom, there are two buttons: "Download Report (PDF)" and "Go Back to Dashboard".

Figure. B.4. Result page

TB Prediction Report

Patient Name: Santhosh
Age: 20
Blood Group: O

TB Status: TB Detected
TB Type: Active TB

Medical Explanation:

This result indicates that the individual has been diagnosed with tuberculosis (TB) and that it is currently active in their body. The risk score of 117.5 suggests that the TB is at a relatively high level of severity. It is important for the individual to seek medical treatment as soon as possible to manage and treat the TB infection.

GradCAM Visualization:

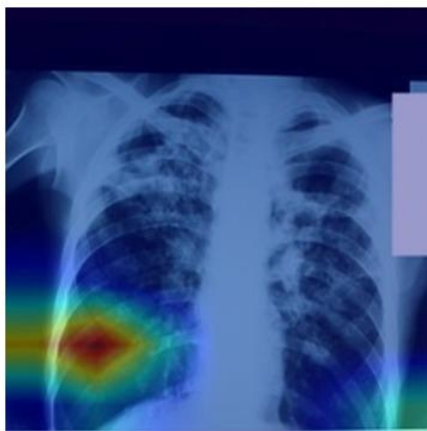


Figure. B.5. Report Generation page

Admin Panel							
TOTAL REGISTERED USERS 58							
Patient Prediction History							
USER	PATIENT NAME	AGE	CBC	ESR	CRP	RESULT	DATE
karthick	karthick	28	9000.0	8.0	2.0	TB Detected	2026-03-07 17:40:32
karthick	karthick	27	9000.0	8.0	2.0	TB Detected	2026-03-07 17:45:27
karthick	karthick	27	9000.0	8.0	2.0	TB Detected	2026-03-07 17:49:08
karthick	karthick	27	9000.0	8.0	2.0	TB Detected	2026-03-07 17:49:14
karthick	karthick	29	9000.0	8.0	2.0	Normal	2026-03-07 17:51:09
karthick	karthick	55	9000.0	8.0	2.0	Normal	2026-03-07 18:06:22
karthick	karthick	54	15000.0	60.0	30.0	TB Detected	2026-03-07 18:26:01
karthick	karthick	29	9000.0	8.0	2.0	Normal	2026-03-07 18:34:50
karthick	Ranjith	28	9000.0	8.0	2.0	Normal	2026-03-07 18:55:52
karthick	karthick	27	9000.0	8.0	2.0	TB Detected	2026-03-10 07:24:43
karthick	karthick	27	9000.0	8.0	2.0	TB Detected	2026-03-10 08:26:35
karthick	karthick	27	9000.0	8.0	2.0	Normal	2026-03-10 08:28:19

Figure. B.6. Patient History Prediction page

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in **SIXTH INTERNATIONAL CONFERENCE ON ARTIFICIAL INTELLIGENCE AND SMART COMPUTING (ICAISC'26)** held during 19th - 21st March 2026 at Bannari Amman Institute of Technology, Sathyamangalam.


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